

①

Method:

$$\text{Start from } \pi(t) \xleftrightarrow{F} \text{sinc}\left(\frac{\omega}{2\pi}\right)$$

Apply duality to get:

$$\text{sinc}\left(\frac{t}{2\pi}\right) \xleftrightarrow{F} ??$$

Apply time scaling to get:

$$\text{sinc}\left(\frac{t}{T_0}\right) \xleftrightarrow{F} ??$$

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$$\text{Use the facts: } \text{sgn}(t) \xleftrightarrow{F} \frac{2}{j\omega},$$

$$u(t) = \frac{\text{sgn}(t) + 1}{2}$$

and the linearity of Fourier Transform to get

$$u(t) \xleftrightarrow{F} \frac{1}{j\omega} + \pi \delta(\omega)$$

②

$$\text{Apply duality to } \text{sgn}(t) \xleftrightarrow{F} \frac{2}{j\omega}$$

$$\frac{2}{jt} \xleftrightarrow{F} 2\pi \cdot \text{sgn}(-\omega)$$

$$\Rightarrow \frac{1}{t} \xleftrightarrow{F} j\pi \cdot \text{sgn}(-\omega)$$

$$\Rightarrow \frac{1}{\pi t} \xleftrightarrow{F} j \text{sgn}(-\omega)$$

(3)

(a)

$$y(t) = x(t) * \frac{1}{\pi t}$$

$$\Rightarrow Y(\omega) = X(\omega) \cdot j \operatorname{sgn}(-\omega)$$

$$\begin{aligned} \text{(i)} \quad |Y(\omega)| &= |X(\omega)| \cdot |j \operatorname{sgn}(-\omega)| \\ &= |X(\omega)| \quad \text{if } \omega \neq 0, \end{aligned}$$

$$\text{(ii)} \quad \angle Y(\omega) = \angle X(\omega) + \angle j \operatorname{sgn}(-\omega)$$

$$= \begin{cases} \angle X(\omega) - \pi/2 & \text{if } \omega > 0 \\ \angle X(\omega) + \pi/2 & \text{if } \omega < 0 \end{cases}$$

(b)

$$\text{(i)} \quad x(t) = e^{j\omega_0 t}$$

$$X(\omega) = \delta(\omega - \omega_0) \cdot 2\pi$$

$$Y(\omega) = 2\pi \delta(\omega - \omega_0) \cdot j \operatorname{sgn}(-\omega)$$

$$= 2\pi \cdot \delta(\omega - \omega_0) \cdot j \operatorname{sgn}(-\omega_0)$$

$$= \begin{cases} -j \cdot 2\pi \cdot \delta(\omega - \omega_0) & \text{if } \omega_0 > 0 \\ +j \cdot 2\pi \cdot \delta(\omega - \omega_0) & \text{if } \omega_0 < 0 \end{cases}$$

$$\therefore y(t) = \begin{cases} -j e^{j\omega_0 t} & \text{if } \omega_0 > 0 \\ +j e^{j\omega_0 t} & \text{if } \omega_0 < 0 \end{cases}$$

(ii) Assume $\omega_0 > 0$. (Similar analysis can be done for $\omega_0 < 0$)

$$\sin \omega_0 t = \frac{e^{j\omega_0 t} - e^{-j\omega_0 t}}{2j}$$

$$\sin \omega_0 t * \frac{1}{\pi t} = \frac{1}{2j} \left[e^{j\omega_0 t} * \frac{1}{\pi t} - e^{-j\omega_0 t} * \frac{1}{\pi t} \right]$$

$$= \frac{1}{2j} \left[-j e^{j\omega_0 t} - j e^{-j\omega_0 t} \right] \quad (\text{from (i)})$$

$$= \frac{-j}{2j} \left[e^{j\omega_0 t} + e^{-j\omega_0 t} \right]$$

$$= -1 \times \left[\frac{e^{j\omega_0 t} + e^{-j\omega_0 t}}{2} \right]$$

$$= -\cos \omega_0 t.$$

For $\omega_0 < 0$, you must get:

$$\sin \omega_0 t * \frac{1}{\pi t} = \cos \omega_0 t$$

Since $\sin(\omega_0 t) = -\sin(-\omega_0 t)$ and $-\omega_0 > 0$,

(iii) Final answer for $\cos \omega_0 t$

$$\cos \omega_0 t * \frac{1}{\pi t} = \sin \omega_0 t, \text{ if } \omega_0 > 0$$

$$\cos \omega_0 t * \frac{1}{T} = -\sin \omega_0 t \text{ if } \omega_0 < 0.$$

④ Apply Derivative properties.

⑤ (a) $f_n(t) = g_n(t) * \sum_k \delta(t - kT_0)$ → don't use 'n' here!

$$= \sum_k g_n(t - kT_0)$$

(b) $F_n(\omega) = G_n(\omega) \times F\left[\sum_k \delta(t - kT_0)\right]$

$$= G_n(\omega) \times \frac{2\pi}{T_0} \sum_k \delta(\omega - k\omega_0) \quad ; \quad \omega_0 = \frac{2\pi}{T_0}$$

$$= \frac{2\pi}{T_0} \sum_k G_n(k\omega_0) \cdot \delta(\omega - k\omega_0).$$

To find $G_n(\omega)$: Note that $g_n(t) = \sqrt{\frac{n}{2\pi}} \cdot \exp(-\frac{nt^2}{2})$.

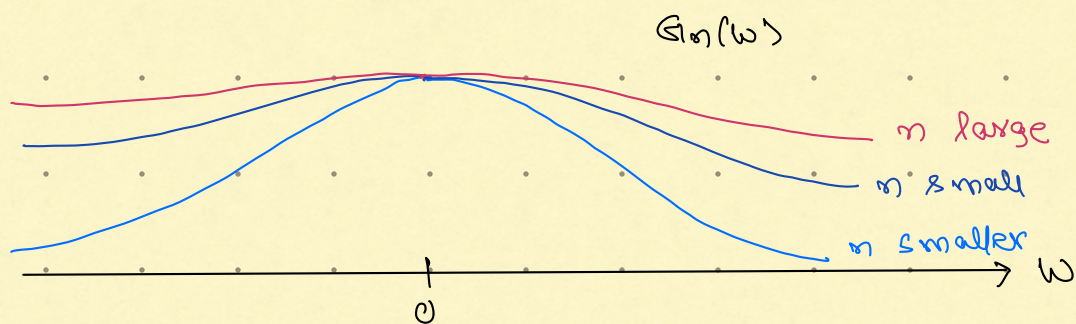
We know: $e^{-t^2} \xleftrightarrow{F} \sqrt{\pi} \cdot e^{-\omega^2/4}$.

Time scaling: $t \rightarrow at$ with $a = \sqrt{n}/2$

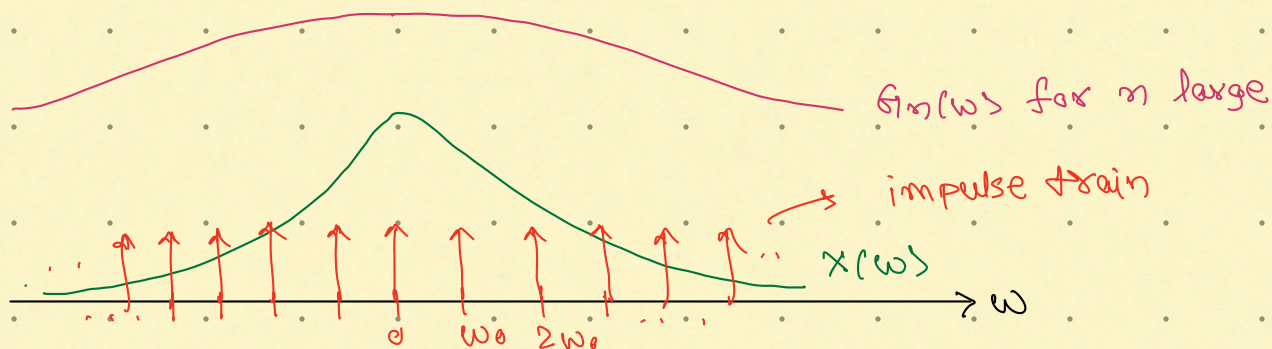
$$e^{-nt^2/2} \xleftrightarrow{F} \frac{\sqrt{\pi}}{\sqrt{n}/2} \cdot e^{-\frac{1}{4} \left(\omega \cdot \sqrt{\frac{2}{n}}\right)^2}$$

$$= \sqrt{\frac{2\pi}{n}} \cdot e^{-\omega^2/2n}.$$

$$\therefore \sqrt{\frac{n}{2\pi}} \cdot e^{-nt^2/2} \xleftrightarrow{F} \exp\left(-\frac{\omega^2}{2n}\right).$$



Note that for any w , $\lim_{n \rightarrow \infty} G_n(w) = 1$.



$$F_n(w) = \frac{2\pi}{T_0} \sum_k \exp\left[-\frac{k^2 w_0^2}{2n}\right] \cdot \delta(w - k w_0)$$

$$\int_{-\infty}^{+\infty} F_n(w) X(w) dw = \int_{-\infty}^{+\infty} \frac{2\pi}{T_0} \sum_k e^{-k^2 w_0^2 / 2n} \delta(w - k w_0) \cdot X(w) dw$$

$$= \int_{-\infty}^{+\infty} \frac{2\pi}{T_0} \sum_k X(k w_0) e^{-k^2 w_0^2 / 2n} \delta(w - k w_0) dw$$

$$= \sum_k \frac{2\pi}{T_0} X(k w_0) \cdot e^{-k^2 w_0^2 / 2n}$$

Now, $\lim_{n \rightarrow \infty} \left[\int F_n(\omega) x(\omega) d\omega \right]$

$$= \lim_{n \rightarrow \infty} \left[\sum_k \frac{2\pi}{T_0} \cdot X(k\omega_0) \cdot e^{-k^2 \omega_0^2 / n} \right]$$

$$= \sum_k \frac{2\pi}{T_0} X(k\omega_0) \left(\lim_{n \rightarrow \infty} e^{-k^2 \omega_0^2 / n} \right)$$

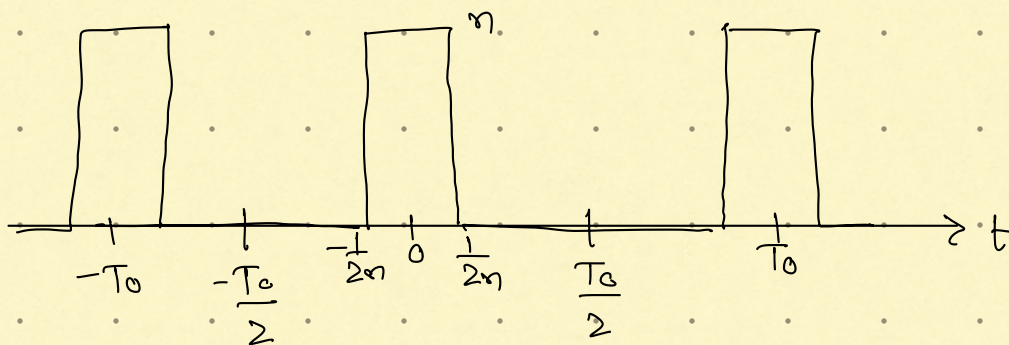
(interchange of $\sum$ and $\lim$ is possible

here; this can be formally proven using the fact $X(\omega)$ is Schwartz)

$$= \sum_k \frac{2\pi}{T_0} X(k\omega_0)$$

⑥

$f_n(t)$ is periodic with period T_0



To find $F_n(\omega)$:

Find $f_n(t) \xleftrightarrow{\text{F.S.}} c_k$

use $F_n(\omega) = \sum_k c_k \cdot \delta(\omega - k\omega_0)$

Rest is similar to question 5.

$$\textcircled{7} \quad T_0 \langle T_2, x \rangle = \left\langle \sum_k e^{jk\omega_0 t}, x \right\rangle$$

$$= \sum_k \langle e^{jk\omega_0 t}, x \rangle$$

$$= \sum_k \int_{-\infty}^{+\infty} x(t) \cdot e^{jk\omega_0 t} dt$$

$$= \sum_k X(k\omega_0)$$

$$\therefore \langle T_2, x \rangle = \frac{1}{T_0} \sum_k X(k\omega_0)$$

$$\langle T_1, x_1 \rangle = \sum_n x_1(nT_0)$$

$$= \sum_n x(-nT_0)$$

$$= \langle T, x_1(-t) \rangle$$

$$= \frac{1}{2\pi} \langle \hat{T}, x \rangle$$

$$= \frac{1}{2\pi} \left\langle \sum_k \frac{2\pi}{T_0} \delta(\omega - k\omega_0), x \right\rangle$$

$$= \frac{1}{T_0} \sum_k X(k\omega_0)$$

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Use time-shift formula to find

$$\text{sinc}\left(\frac{t - nT_0}{T_0}\right) \xleftrightarrow{F} ??$$

and then apply Plancherel's formula.

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$$y(t) = \int_{-\infty}^{+\infty} \frac{1}{a} \text{sinc}\left(\frac{\tau}{a}\right) \cdot \frac{1}{b} \text{sinc}\left(\frac{t-\tau}{b}\right) d\tau$$

$$= \int_{-\infty}^{+\infty} \frac{1}{a} \text{sinc}\left(\frac{\tau}{a}\right) \cdot \frac{1}{b} \text{sinc}\left(\frac{\tau-t}{b}\right) d\tau$$

$$(\text{sinc}(\text{sinc}(-x)) = \text{sinc}(x))$$

$$= \int_{-\infty}^{+\infty} \frac{1}{a} \text{sinc}\left(\frac{\tau}{a}\right) \cdot \frac{1}{b} \text{sinc}^*\left(\frac{\tau-t}{b}\right) d\tau$$

Now apply Plancherel's theorem.

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(a)

$$H_{\text{HPF}}(\omega) = 1 - H_{\text{LPF}}(\omega)$$

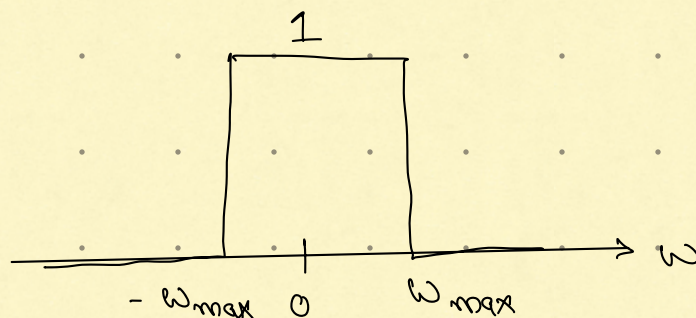
$$\Rightarrow h_{\text{HPF}}(t) = \delta(t) - h_{\text{LPF}}(t)$$

Find $h_{\text{LPF}}(t)$ and use it here.

(b) Let $\omega_{\max} = \frac{\omega_1 - \omega_0}{2}$

and $\omega_c = \frac{\omega_1 + \omega_0}{2}$

Consider $H_{LPF}(\omega)$:



Then, $H_{BPF}(\omega) = 2 H_{LPF}(\omega) * \left[\frac{\delta(\omega - \omega_c) + \delta(\omega + \omega_c)}{2} \right]$

$\Rightarrow h_{BPF}(t) = 2 h_{LPF}(t) \cdot \cos(\omega_c t)$

Substitute the value of $h_{LPF}(t)$ here.

(13) First why: because of convolution property

Second why: $\langle T(t), x(t) \rangle = \int_{-\infty}^{+\infty} T(t) \cdot x(-t) dt$

$= \lim_{n \rightarrow \infty} \left[\int_{-\infty}^{+\infty} f_n(t) \cdot x(-t) dt \right]$

$= \lim_{n \rightarrow \infty} [\text{value of } f_n * x \text{ at } t=0]$